Chapter 24

Ethics in Clinical Research

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Learning Objectives:

Objective 24.1 Overview untoward drug incidents that influenced the evolution of ethics in clinical research.
Objective 24.2 Explain salient features of laws and guidelines in history to counter unethical research.
Objective 24.3 Detail the constitution and process of review by the research ethics committees.
Objective 24.4 Describe the informed consent process in clinical research.
Objective 24.5 Examples of some unethical practices in current clinical research.

OBJECTIVE 24.1. OVERVIEW UNTOWARD DRUG INCIDENTS THAT INFLUENCED THE EVOLUTION OF ETHICS IN CLINICAL RESEARCH

The term "ethics" was derived from the Greek word meaning "Ethos"[1] (Fig. 24.1). Clinical research should be conducted ethically to protect the rights, safety, and well-being of the study participants.[2] Pharmacists have responsibilities to contribute to the ethical conduct of research no matter where they practice. Training in biomedical research ethics is mandatory for many institutions for their health professionals involved in clinical research. The term Ethos was used to describe the guiding beliefs (norms or customs) of the people. As a guardian of providing safe and cost-effective medications or pharmacy services to consumers, pharmacists must lead in the ethical conduct of clinical research. For many, ethics is a philosophical matter and may be considered as subjective and vary from person to person based on their perceptions or may be relative in nature varying from situation to situation.[3]

Clinical research involves studies conducted in healthy volunteers or patients. There are times when the science of the research might conflict with the well-being of the research subject. The ultimate question then is which is more important, science or ethics? In clinical research, the answer is simple, ethics is the first consideration. The rights of a few should not be sacrificed for the benefit of others. It does not mean that scientific quality in research is not important because poor methodological quality will result in questionable results rendering the study efforts a waste. However, unnecessary exposure of participants to any stimulus (whether it is a drug or a service) is unethical. Thus nonscientific methods are unethical.[4] Therefore, it is the responsibility of an ethics committee to review both the ethical and scientific aspects of a research proposal.[5]

Research has not always been conducted with high ethical standards. Clinical research must always be socially responsible.[6] Historical examples of unethical research practices that focused more on generating scientific knowledge and not on subject safety have led to important changes in how research is conducted today. Other examples of inappropriate conflicts of interest have also forced the development of ethical guidelines and related regulatory requirements. Studying examples of unethical practices from the past is vital to get a thorough understanding of modern ethical practices. The following are summaries of some well-known unethical clinical research from history.

Tuskegee Syphilis Study (1932–72)

Almost 600 underprivileged men (mostly African–American) constituted the Tuskegee Syphilis Study, one of the most notorious of unethical research studies ever conducted.[7] There were even attempts to rationalize the research once it was

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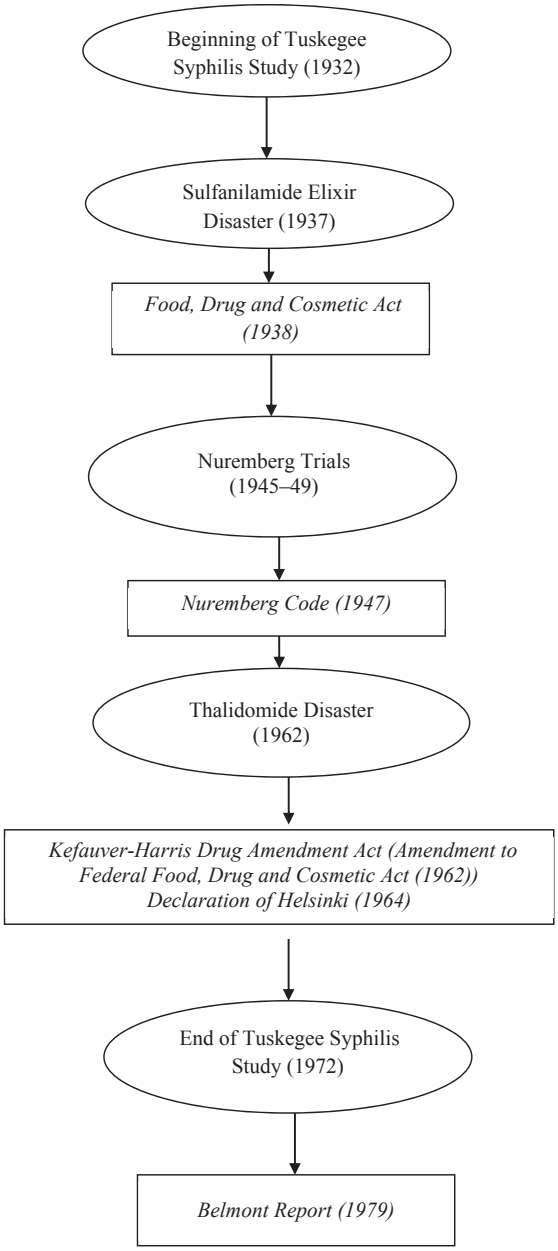


FIGURE 24.1 Evolution of biomedical ethics.

criticized for its unethical nature.[8] It was a United States Public Health Service prospective study in Macon County, Alabama, to study the "natural history" of syphilis. In the trial, around 399 of the participants were infected with syphilis (a sexually transmitted disease), and 201 participants who did not have the disease served as control. The trial was conducted over a period of 40 years, and the study evaluated the progression and complications associated with the untreated disease. The researchers had not explained the purpose of the trial nor treated the participants, though penicillin was available as the recommended drug of treatment for syphilis. It was in 1972 that the study was abandoned at the insistence of the press. By then many of the subjects had died because of the disease and its complications. In addition, a few had transmitted the infection to their spouses, and their children were born with congenital syphilis.[9] The ethical issues identified were as follows:

- Lack of informed consent. Study participants were not told specifically that the study aimed to generate knowledge on syphilis progression. Participants were deceived that they were treated while it was just diagnostic procedures.

- Penicillin was available, but the study participants were not provided with it even though it could cure syphilis. The study protocol prevented participants from receiving penicillin.
- No records were maintained. The actual number of participants who died due to syphilis was never known.
- Although parts of the study appeared in medical journals for many years, the study was not stopped until news in the national press appeared in 1972.[8,10]

Elixir Sulfanilamide Disaster (1937)

The Elixir Sulfanilamide disaster of 1937 was one of the most consequential mass poisonings of the 20th century. Sulfanilamide, an antibacterial drug, was being used safely in the treatment of streptococcal infections. A new elixir formulation with raspberry flavor was prepared using 70% diethylene glycol (DEG) as there was a need for the drug in its liquid form. DEG can be fatal, and unknowingly the company distributed the product across the United States. However, the Food and Drug Regulation (at that time) did not insist on toxicity studies for the new formulation. Hence, the company did not carry out any toxicity studies. Soon after its appearance on the market, the American Medical Association was made aware of the lethal cases from the administration of this elixir, and though immediate attempts were made to recall it, many died. The elixir was produced by a US pharmaceutical company, S. E. Massengill Company. The formulation resulted in the death of 105 patients who consumed the elixir. In reaction to this calamity, the US Congress passed the 1938 Federal Food, Drug and Cosmetic (FD&C) Act, which required proof of safety before the release of a new drug. The 1938 law changed the drug focus of the Food and Drug Administration (FDA).[11,12]

Nuremberg Trials (Nazi Trials 1945–49)

There were several unethical experiments conducted on the prisoners of war in Germany during World War II. Prisoners endured intolerable treatments as they fell victim to many experiments conducted by Nazi doctors.

Numerous inhumane experiments under the disguise of medical research were identified in the Nuremberg war trials[13]:

- Gypsy twins were targeted to understand their genetics and were then brutally killed with phenol injections. Autopsies were performed to compare the similarities among their internal organs. Different chemicals were injected into them to study the color change of their irises.
- Bones were removed and transplanted in prisoners without anesthesia. No concern was given to prisoners who suffered pus infections after the surgery. Some of them developed permanent disabilities after such experiments.
- Jewish females were exposed to a chemical irritant and were mass sterilized leading to dysfunction of the reproductive system. Subsequently, females, as well as male prisoners, were exposed to high intensity X-ray radiation on their reproductive organs.
- High-altitude experiments were carried out to mimic the conditions of altitudes up to 60,000 feet. Such high altitudes were chosen to study the low-pressure limits required for pilots to eject during an emergency. Such low-pressure chambers lacked oxygen, which often led to the death of the victims. On the survivors, the Nazis performed experimentation on the brain (vivisection).
- Many individuals were intentionally burned to see how the wounds would heal.
- Some individuals were dissected while alive so that anatomical drawings of internal organs could be done, and these were published in an anatomy textbook.[13]

Cutter Incident (1955)

The Cutter incident happened in the developmental struggle of a polio vaccine. In April 1955, 10 children died and 200 became paralyzed to varying degrees after receiving polio vaccine in the United States. Cutter Laboratories made the vaccine using Salk's formaldehyde-treatment method that was found to be defective. This incident led to adoption of Sabin's vaccine using an attenuated strain of the virus orally. There were also incidences of polio developed with Sabin's vaccine, and then a different version of Salk's vaccine was marketed. Current strict measures provide better safety profiles of vaccines but are only being produced by a few manufacturers. This has led to vaccine shortages.[14]

Thalidomide Disaster (1962)

Thalidomide is a nonbarbiturate sedative for treating insomnia and was originally marketed by a German manufacturer. From 1961 to 1962, the drug was also used in pregnant women for morning sickness. It was marketed as a

safe drug, but about 10,000 cases of phocomelia (shortening or absence of limbs) were reported, and about half of the fetuses died during pregnancy. The disaster occurred mainly in Europe, Japan, and Russia. The drug was not approved in the United States due to lack of safety information. This tragedy led to major concerns about the use of medications in pregnancy and the concept of the fetus as a vulnerable subject.[12,15] Thalidomide is currently available in the global market for the treatment of cancer with an absolute contraindication in pregnancy or women of childbearing age.

The Unfortunate Experiment (1960s–80s)

This was a natural history study of cervical intraepithelial neoplasia 3 in women at the National Women's Hospital, Auckland, New Zealand. Available treatments for the disease were withheld from the patients. The study had an ethics committee approval; however, based on current guidelines the study would be considered wrong factually and ethically. Women with carcinoma in situ (CIS) of the cervix were followed, but not treated, to prove that CIS is not a premalignant disease. The outcomes for women with microinvasive cancer had not previously been reported at the time of the study. More than 100 women with CIS and microinvasive (Stage 1A) cancer of the cervix, vagina, and vulva were not treated for varying lengths of time.[8] These conditions have a substantial risk of developing into higher stages of invasive cancer. This study is popularly known as the "Unfortunate Experiment."[16]

Other Major Issues Reported in Recent Times

Conflict of interest is a major issue affecting ethics in research and drug development. Researchers are responsible for conducting all research experiments with strict ethical standards, including the following: (1) design of the research; (2) fair enrollment of subjects; (3) analysis of results; and (4) the presentation of results. For pharmaceutical industries, the financial benefits of getting a drug approved for commercial use is a leading conflict of interest. For research professionals, receiving payments for conducting studies, paid advocacy for new drugs, and lucrative payments to be consultants are all major conflicts of interest. Other common reported ethical issues are as follows:

- Pharmaceutical companies are influencing health professionals to perform unnecessary research and/or prescribe their drugs.
- Researchers manipulating study results to be published to or remain in the good graces of commercial funders of their research.
- Hiding (failing to publish) negative study results.
- Aggressive marketing by companies of uses that are outside the approved indications of drugs.

OBJECTIVE 24.2. EXPLAIN SALIENT FEATURES OF LAWS AND GUIDELINES IN HISTORY TO COUNTER UNETHICAL RESEARCH

Regulatory initiatives and changes evolved with lessons learned from drug tragedies.

Food, Drug and Cosmetic Act (1938)

After the grief of the Sulfanilamide Elixir tragedy, the United States Congress passed the FD&C Act in June 1938. This legislation created the FDA and required that manufacturers have to apply to the FDA before any new drug could be marketed.[11]

Nuremberg Code (1947)

The cardinal principles of Nuremberg Code can be categorized into three areas[17]:

1. Subject related
2. Investigator related
3. Experiment related

Ten points under these three categories are illustrated in Fig. 24.2.

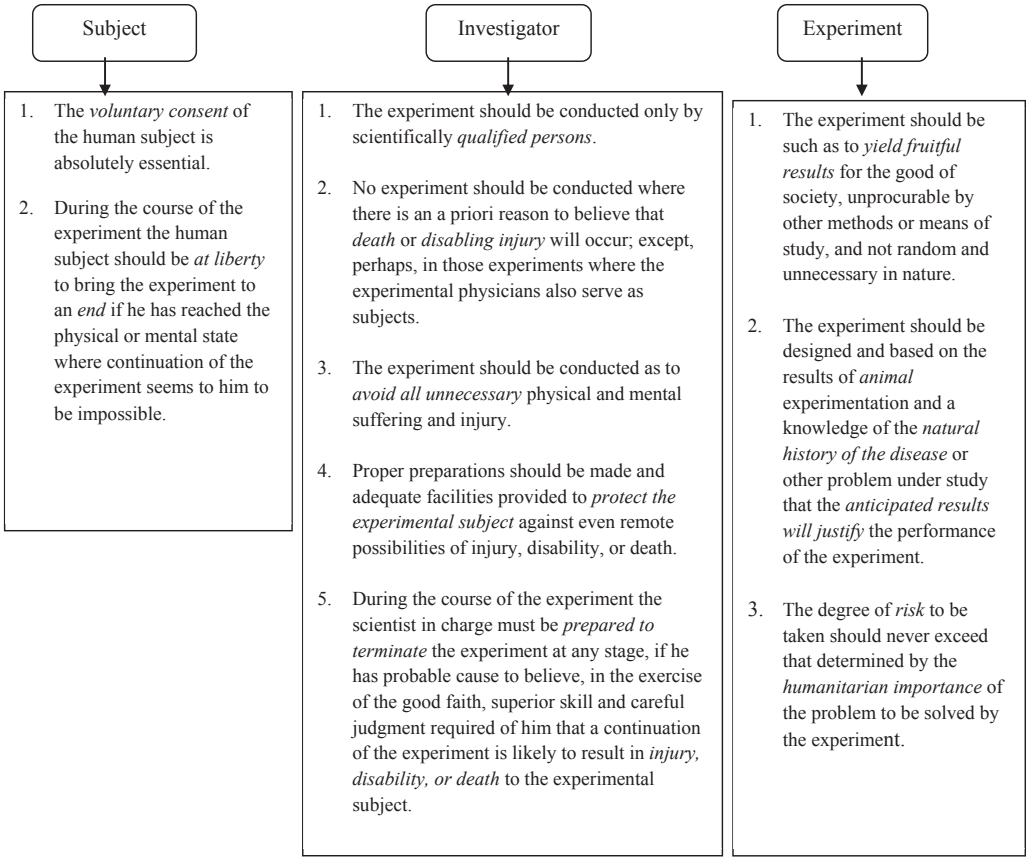


FIGURE 24.2 Cardinal principles of the Nuremberg Code.

The Nuremberg Code provided basic moral, ethical, and legal principles for conducting research, but the problem was that there was no way to enforce these principles. The regulations that established the FDA were able to address the sulfanilamide disaster but did not address the issues related to research for changes to approved drugs undergoing formulation changes.

Kefauver Harris Drug Amendment Act 1962 (Amendment to Federal Food, Drug and Cosmetic Act)

The amendment named after Senator Estes Kefauver states that drug manufacturers must prove the efficacy of drugs and their various formulations before they can be marketed and will report any serious side effects that have been determined. The efficacy of the drug being evaluated should be from subjects willing to consent to participate in adequate and well-controlled trials conducted by expert physicians. It also mentioned that the FDA should approve the marketing application before the drug could be marketed. This amendment was able to address the concerns of research on new drugs and formulations.[18]

Declaration of Helsinki

The World Medical Association met in Helsinki, Finland, in 1964 and developed the primary ethical guidelines for studies involving human beings. Finally, 16 years after the Nuremberg Code, there was a way that made physicians ethically bound to adhere to these principles. This Declaration of Helsinki has been followed by many revisions with the most recent Revision 9 being released in 2013.[19] It is widely used in current practice. Many other guidelines directed at the ethical conduct of research are based on the Declaration of Helsinki.

General Principles

The physician must consider the health of his/her patient as the foremost consideration. It is his/her duty to promote and safeguard the health, well-being, and rights of the patient involved in medical research (Figs. 24.3 and 24.4).

The group of people (vulnerable) in Fig. 24.5 can be involved in medical research, only if a proper justification has been provided for trials that cannot be conducted in nonvulnerable groups. In addition, a trial may have its priority on vulnerable people by improvement of their health condition.

Scientific Requirements and Research Protocols

A research protocol must have adequate details scientifically (trial design and methodology), technically (trial execution), and ethically (safety and well-being of the subjects).

Research Ethics Committees

The committee must be independent of the researcher and should have enough qualified persons to approve the research protocol after a thorough review and should ensure the execution and documentation of the trial that it complies with the protocol and regulatory requirements. The researcher should submit periodic reports regarding the progression of the clinical trials, interim monitoring reports that include serious adverse events (if any), and also a final report to the committee at the end of the study.

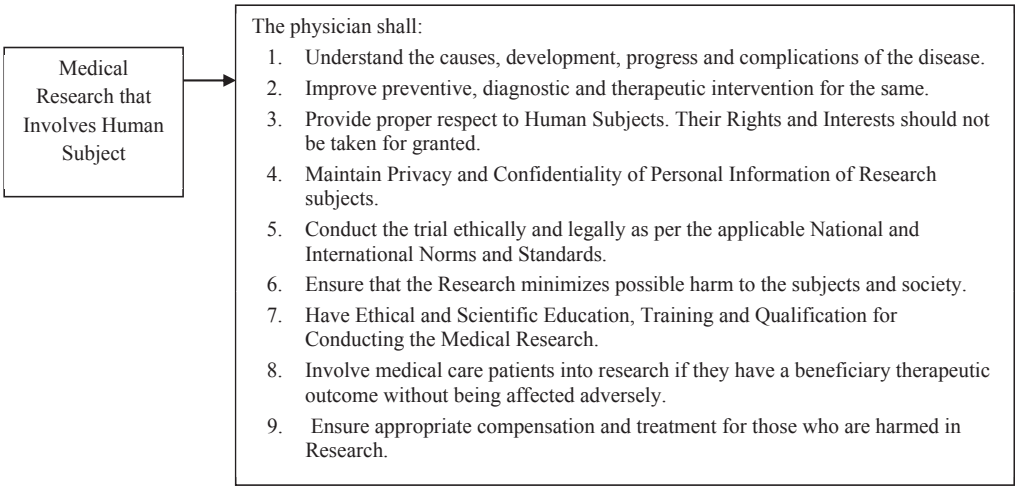


FIGURE 24.3 Principles of the Declaration of Helsinki (2013 version).

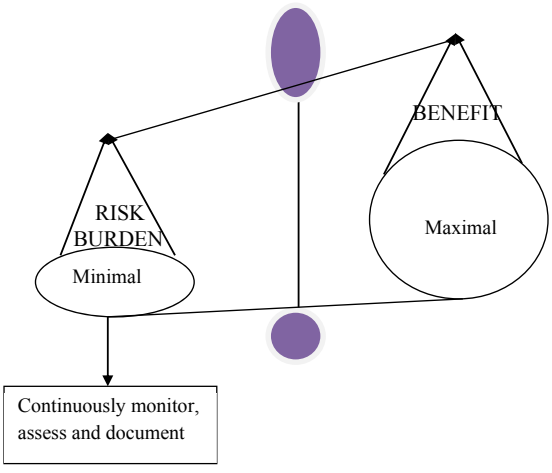


FIGURE 24.4 Risks, burdens, and benefits of clinical research.

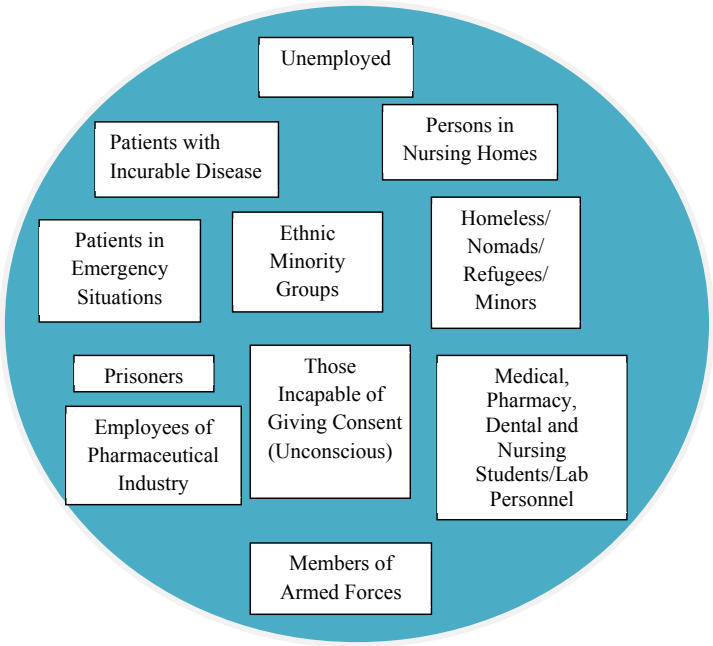


FIGURE 24.5 Vulnerable groups and individuals.

Privacy and Confidentiality

The privacy of the subjects and confidentiality of their medical records must be maintained.

Informed Consent

Subjects should be informed of the elements of informed consent during the process such as objectives, methodologies, details about the trial medications, alternative treatments available, risk/benefit ratio, compensation, insurance policy, etc. The physician/research team shall not influence the participation of the subjects in the trial if he/she has a dependent relationship with the subjects. In such cases, informed consent should be sought by another qualified individual who is independent of this relationship. Similarly, a refusal to participate should not adversely affect the healthcare professional–patient relationship.

Use of Placebo

A placebo shall be used when there is no proven intervention. In addition, in cases of any new intervention that is less efficacious than the best-proven treatment, the use of placebo is necessary to determine the efficacy and safety of that intervention.

In the case of intervention, which is less efficacious when compared to the best-proven treatment, a placebo may again be used as it will not produce any serious risk or harm, as a result of not receiving the best-proven intervention.

Posttrial Provisions

Sponsors, researchers, and government shall allow posttrial provisions for all participants who still need an intervention identified as beneficial in the trial.

Research Registration and Publication and Dissemination of Results

Studies involving humans should be registered in a publicly available database before recruitment of the first subject. All results that are positive, negative, and also inconclusive must be published or otherwise be made publicly available. Periodic updates of research outcomes should be made available to the public by updating the clinical trial registries or databases.

Unproven Intervention in Clinical Practice

In any treatment, when proven interventions do not exist, and other known interventions are ineffective, then an unproven intervention may be used after obtaining informed consent from the subject or his/her legally authorized representative. Such unproven interventional trial may be carried out when the expected outcome is the hope of saving the life of the subject, reestablishing health, or alleviating the suffering of the subject. This unproven intervention can be subsequently involved in research to evaluate its efficacy and safety.[19]

The Declaration of Helsinki provided guidelines about the safe and ethical conduct of research. Members of the World Medical Association required its members to adhere to these basic principles of research. So there was now a mechanism in place to ensure good quality research. However, the Tuskegee exposé and other public reports revealed that additional changes were required. The United States responded with the National Research Act of the US Congress, 1974. This legislation established the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research (the Belmont Commission) and charged the Commission to

- identify the basic ethical principles for research,
- develop guidelines for research,
- make recommendations about how to implement these changes.[20]

Belmont Report (April 18, 1979)

The Belmont Report defined core ethical principles.[38] The three cardinal principles of biomedical ethics identified in the report are: (1) respect for persons, (2) beneficence, and (3) justice, which are fundamental to understanding current ethical practices.[21]

Respect for Persons: categorizes people into two groups "autonomous individuals" and "individuals with diminished autonomy." Autonomy is respecting competent people for their abilities to take a decision. It is also protecting the autonomy of the vulnerable (e.g., prisoners, psychiatric patients) by preventing the imposition of unwanted decisions. The consent process protects autonomy.[22]

Autonomous individuals are those who are capable of self-thinking and can act individually. Their decision of voluntariness and willingness to participate in a trial should be respected.

Individuals with diminished autonomy are a vulnerable group of the population who seek protection from activities that are harmful, and that would adversely affect them. These groups should not be unduly influenced by participation and should also be respected.[20]

Beneficence is an obligation to provide benefits and to balance benefits against risks.[23] Nonmaleficence means the participants should not be harmed through the conduct of the study.[22] Maximize possible benefits (beneficence) and minimize possible harms (nonmaleficence) to the trial participants. The benefit of participating in a trial is the chance to receive a treatment that may not be available in the market for many years. With randomization, it is unsure whether the participant will receive the expected treatment. To manage this ethical problem, crossover designs allow the placebo group and the treatment group to experience both and are encouraged when the outcome allows this crossover.

No intervention with an expected benefit is without risks. In such cases, the degree of risk determines the continuity or the progress of the trial. Risks should always be weighed against the possible benefits of participating in the clinical trial. The principle of beneficence is to minimize risk or avoid it to the maximum extent possible, thereby protecting the physical, mental, and social well-being of the subjects.

Justice in clinical research demands that all subjects be treated fairly. Thus, equals must be treated equally.[22] Justice refers to the impartial distribution of risk and benefits of participation to different groups of subjects. Recruitments of subjects should also be performed impartially. The vulnerable population should never be coerced or exploited for participation in a trial that would be highly unfavorable regarding justice. Equality in human subjects participating in clinical research is to be ensured.[24] However, not all human subjects are the same; people differ with respect to age, health, gender, race, mental abilities, socioeconomic status, and other characteristics. The subjects with similarities could question if they receive unequal privileges.[24]

Respect for Persons

- Principles of voluntariness, informed consent, and community agreement: Risks and benefits are clear to participants and the decisions are individual.
- Principles of nonexploitation: Appropriate compensation and assured medical treatment for potential harms.
- Principles of privacy and confidentiality: Except when required for legal reasons, personal information should not be disclosed.

Beneficence

- Principles of essentiality: Is it essential to conduct research?
- Principles of precaution and risk minimization: Prevent and minimize risk, the trial should be stopped if required.
- Principles of professional competence: The research team should act with integrity and impartiality.
- Principles of accountability and transparency: Record actions and avoid conflicts of interest.

Justice

- Principles of the maximization of the public interest and distributive justice: Research should be of benefit to society.
- Principles of the public domain: Research findings to be shared with the body of scientific knowledge through appropriate mediums of communication.
- Principles of totality of responsibility: Be responsible in general.

International Council on Harmonization and Good Clinical Practice

To streamline the process of clinical trials, different countries have drafted and finalized their own set of guidelines and laws. When a pharmaceutical company plans to market a drug in different countries, they find it difficult to adopt different regulatory guidelines for drug product registration. To avoid these discrepancies among different drug regulatory agencies across the globe, the European Union, the United States, and Japan initiated a harmonization procedure in consultation with the International Federation of Pharmaceutical Manufacturers and Associations. The regulatory personnel of the European Union, the United States, and Japan planned several international conferences, and in 1990 the International Council on Harmonization (ICH) was founded in Brussels.[2]

In 2017, the following drug regulatory agencies were members of the ICH[26]:

- European Union
- The United States FDA
- Ministry of Health, Labour, and Welfare/Pharmaceuticals and Medical Devices Agency, Japan
- Health Canada, Canada
- Swissmedic, Switzerland
- Ministry of Food and Drug Safety, Republic of Korea
- Agência Nacional de Vigilância Sanitária, Brazil
- China FDA, China

The ICH has released guidelines in the following four categories[27]:

Q—Quality
Harmonization achievements in the quality area include pivotal milestones such as the conduct of stability studies, defining relevant thresholds for impurities testing and a more flexible approach to pharmaceutical quality based on Good Manufacturing Practice (GMP) risk management.

S—Safety
ICH has produced a comprehensive set of safety guidelines to uncover potential risks like carcinogenicity, genotoxicity, and reproductive toxicity. A recent breakthrough has been a nonclinical testing strategy for assessing QT interval prolongation liability, the single most important cause of drug withdrawals in recent years.

E—Efficacy
The work carried out by ICH under the efficacy heading is concerned with the design, conduct, safety, and reporting of clinical trials. It also covers novel types of drugs derived from biotechnological processes and the use of pharmacogenetics/genomics techniques to produce better-targeted drugs.

E 6. Good Clinical Practice
ICH enacted Good Clinical Practice (GCP) Guidelines version 1 in 1996. GCP describes the roles and responsibilities of investigators, sponsors, monitors, and the Institutional Review Board (IRB)/Independent Ethics Committee (IEC). It details specific aspects of how clinical trials need to be conducted, the elements of the informed consent document, protocol, and essential documents for the conduct of a clinical trial. A finalized addendum to revision 2 was adopted in November 2016.[28]

M—Multidisciplinary
These are the cross-cutting topics that do not fit uniquely into the quality, safety, and efficacy categories. It includes the ICH medical terminology, the Common Technical Document, and the development of Electronic Standards for the Transfer of Regulatory Information.

Guidance From the Council for International Organization of Medical Sciences in Collaboration With the World Health Organization

Similar to the ICH, the **Council for International Organizations of Medical Sciences** (**CIOMS**) is an international nongovernmental organization established jointly by World Health Organization (WHO) and UNESCO in 1949. Its members include countries like Belgium, South Africa, Norway, Israel, Republic of Korea, and a few other European countries. CIOMS serves to unify the common ethical guideline interests of the international biomedical community. In 1993, CIOMS released the *International Ethical Guidelines for Biomedical Research Involving Human Subjects* and then updated it in 2002 and 2016. These guidelines relate mainly to ethical justification and scientific validity of research:

- ethical review;
- informed consent;
- vulnerability of individuals, groups, communities, and populations;
- women as research subjects;
- equity regarding burdens and benefits;
- choice of control in clinical trials;
- confidentiality;
- compensation for injury;
- strengthening of national or local capacity for ethical review; and
- obligations of sponsors to provide healthcare services.[29,30]

OBJECTIVE 24.3. DETAIL THE CONSTITUTION AND PROCESS OF REVIEW BY THE RESEARCH ETHICS COMMITTEE

The Belmont Report and resulting US federal regulations, the ICH, and the CIOMS/WHO publication all require that medical research being conducted on human subjects must be written and must be reviewed, approved, and monitored by an ethics committee.

Ethics committees for research are named differently in some countries, but all follow similar principles of operation. Common names are IRB, IEC, research ethics committee, and ethics and scientific committee. The two most common terms, IEC and IRB, are the cardinal authority to oversee the conduction of clinical research studies in accordance with the guidelines and regulatory requirements.[2]

IEC: As the name suggests, it is an independent body consisting of technical and nontechnical members. The prime responsibility of the committee is to protect the rights of study participants included in clinical research and to ensure their safety and well-being. The committee is also responsible for ensuring that the research is carried out according to applicable regulatory guidance and ethical principles.

IRB: This is formed within the members of an organization/institution that holds prime responsibility as stated above.

Composition of the IEC/IRB

They should be constructed with at least five members:

- At least one member of a scientific area (medical scientist, preferably a pharmacologist).
- At least one clinician.
- A legal expert/judge.
- A layperson from the community.
- Ethicist/logician/philosopher.

The committee should have a minimum of five members and a maximum of eight to ten, equally distributed in gender. Each member of the committee plays a vital role in providing the decision of the protocol bearing in mind the regulatory guidance and standards of practice.

The IEC/IRB should have sufficient members and should be unbiased in determining the approval of the study protocol. Members should be knowledgeable in reviewing the protocol and its related documents and be able to function effectively as a team to identify and analyze technical difficulties of the study, if any, and also to ensure that the ethical aspects remain uncompromised to the participants.

In the case of studies that include a vulnerable population, the IRB/IEC can include one or more individuals as members of the ethics committee who are technically sound in handling and working with such a population.[2]

Registration of the IEC/IRB

In the United States, IRBs should be registered with Health and Human Services, the Office of Human Research Protection, and renewal should be done every 3 years.[31] Other countries also have their respective agencies to register IEC/IRBs.

Documents that Need to Be Submitted to the IEC/IRB for Approval

- Protocol and its amendments if any
- Informed consent documents
- Case report forms and other annexes of protocol
- Investigator's brochure/prescribing information or summary of product characteristics
- Curriculum vitae of investigator(s) justifying their qualifications to conduct that particular research
- Undertaking by investigator
- Financial disclosure of investigator
- Insurance and indemnification policies/compensation in case of any trial-related activities
- Form of brochures or any advertisement(s) through which the subjects could be recruited
- Agreement between investigator(s) and study sponsors

After submission of documents to the IEC/IRB, if necessary, the documents can be modified or changed at any stage; the final version after all amendments made must be submitted to the IEC/IRB as well.[2] The final version with changes should be denoted by date for easy accessibility during the conduct of the trials. Fig. 24.6 provides the major criteria that need attention before submitting to the IEC and during its review.

Communications and Functions of the IEC/IRB

The IEC/IRB should have a standard operating procedure in which the following should be described:

- Names of the members of the IEC with their qualifications and their designations in the committee.
- The frequency of meetings to review and approve protocols and for notifying the sponsor/investigator for initiating the conduct of the study.
- The frequency of interim review of trials.

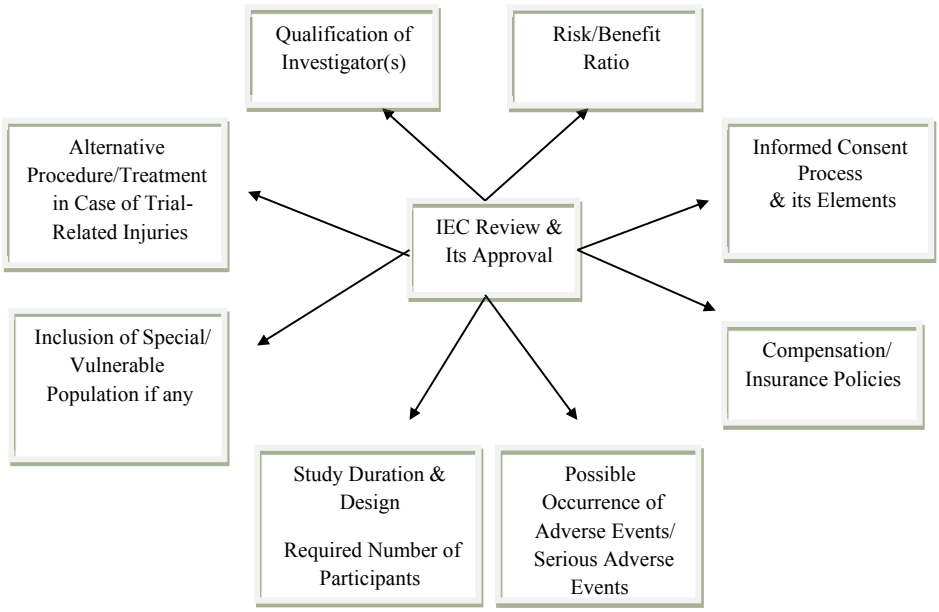


FIGURE 24.6 Ethics committee review of research protocol.

The IEC/IRB must conduct a thorough scrutiny of the informed consent documents. Informed consent communicates risks to subjects, if any, regarding anticipated benefits and the importance of the knowledge that may reasonably be expected to result. The selection of subjects is equitable. Informed consent will be sought from each prospective subject or the subject's legally authorized representative. The consent must be appropriately documented.[2]

The IEC/IRB must ensure that, when appropriate, the research plan makes adequate provision for monitoring the data collected to ensure the safety of subjects. When appropriate, there are adequate provisions to protect the privacy of subjects and to maintain the confidentiality of data. When some or all of the subjects are likely to be vulnerable to coercion or undue influence, such as children, prisoners, pregnant women, mentally disabled persons, or economically or educationally disadvantaged persons, additional safeguards have been included in the study to protect the rights and welfare of these subjects.

A statement must be made about the role of the investigator(s) to notify the IEC/IRB following any serious unexpected event. A statement must also be made that the study would not be conducted unless approved by the IEC/IRB. The IEC/IRB bears the responsibility of conducting spontaneous audits/reviews of ongoing trials at regular intervals and informing the institution/investigator on the actions taken or findings.

Expedited Review Procedure of the IEC/IRB

Expedited reviews may be carried out by the IEC/IRB chairperson or by one or more members of the ethical team designated by the chairperson. Expedited reviews may happen when

1. Research involves minimal risk
2. There is a minor change in an already approved research (which is within a time frame of 1 year or less) protocol.

During such expedited reviews, all functions/authorities of the IEC/IRB may be carried out except for disapproval of research. Disapproval of research may happen only in the nonexpedited review process.

The IEC/IRB may suspend or terminate a research if it is not being conducted as per IEC/IRB requirements or if there is the occurrence of serious unexpected harm to subjects. Such action by the IEC/IRB shall be reported to investigators and regulatory agencies.[2]

OBJECTIVE 24.4. DESCRIBE THE INFORMED CONSENT PROCESS IN CLINICAL RESEARCH

Obtaining informed consent in a trial is one of the most essential steps and should always be a simple process. The informed consent process should be done by extensively detailing all the procedures and risks and benefits involved by participating in the clinical trial. Participants involved in the trial should be respected and never be forced to participate in a trial. Their willingness to participate should be established. The study participants should be given adequate time to understand the consent, and he/she must sign and date the informed consent document. This process should convey the fact that the trial would be conducted in accordance with all the governing regulations, and participation is purely voluntary.

Consent Process

The process should not include the mere signing of the document but should allow a proper interaction between the researcher and the participant. The process should also not coerce any volunteer for their participation. Consent should be presented orally by the investigator. Appropriate time should be given to the subject to read and understand the document.

Aspects such as risk/benefits involved in the trial, the duration of the trial, the occurrence of adverse events due to trial medication, and compensation for participation should be appropriately described in the consent form and discussed. Individual interactions, as well as group discussions, are required. The participant should be free to question the researcher about any trial-related matter. The ability to withdraw at any time from the trial should be known to the participant.

The informed consent document should be easy to understand and technical phrases and words should be avoided. Transparency here defines the occurrence of all adverse events (possibility of risks) and benefits gained from the trial. In case of no study benefits, the same should be mentioned in the informed consent form.

At any point during the trial, whenever new information becomes available that may require a subjects to reconsent, the same information should be included in the revised informed consent form for the IEC/IRB's approval. The subject or his/her legally acceptable representative (LAR) should be informed promptly so that they can decide on their further continuation in the trial.

The following should be explained in the informed consent form[2]:

1. A statement that explains that the trial involves research.
2. The purpose of the conduct of the trial. The duration of the trial (i.e., the subject's expected time of participation in the study, washout period, and randomization procedures).
3. Procedure of the trial (medically invasive procedures, if any, and any other experimental procedures).
4. The foreseeable risks and expected benefits that could happen from the trial.
5. Any alternative procedure/treatment adopted during the course of the trial or during the occurrence of an adverse event(s).
6. Compensation/medical treatments that may be provided in case of trial-related injuries. The expected payment that the subjects would receive and also the expected expenses, which may occur during participation.
7. A confidentiality statement that the records of the subject will be granted direct access only to the ethics committee, sponsors, and regulatory authorities. The records will be kept confidential and will not be made known to the public. The results may be published, but the subject's identity will not be disclosed.
8. A statement that claims participation in the trial is purely voluntary and the subject can withdraw from the trial at any time without penalty or loss of benefits to which the subject is entitled.
9. A statement about whom to contact for trial-related queries and injuries.
10. A statement involving unforeseeable risk to subject and fetus in case of inclusion of pregnant women in the trial.
11. Reasons and circumstances for possible termination of subject's participation by the investigator without his/her consent.
12. Procedure for withdrawal from a study by the subject.
13. An approximate number of subjects participating in the trial.

The statements involving the above elements should be documented and approved by the IEC/IRB. The IEC/IRB is responsible for ensuring the adequacy of the document. The document can then be circulated among the subjects or the subject's LAR.

The document should be provided to the subject in a language that they best prefer. The English version should be translated into vernacular languages for non-English-speaking participants. Utmost care should be taken while translating. The meanings of phrases and sentences should be appropriate, and the similarity between the languages should be maintained.

The document should be signed and dated by the subject or the subject's LAR.[2]

Role of Legally Acceptable Representative

The LAR is an individual or judicial or other body authorized under applicable law to consent on behalf of a prospective subject to the subject's participation in the procedure(s) involved in the research.

An LAR is required for an illiterate person, a pediatric patient, a geriatric patient with diminished cognition, or someone in an unconscious medical condition.

The above subjects can participate in trial interventions with the help of an LAR. The LAR is responsible for ensuring that the subject receives benefits from the trial in a secure way and their rights and interests are protected.[2]

Responsibilities of the Legally Authorized Representative

- The LAR should thoroughly read and explain all the points described in the informed consent to the subject.
- The LAR should not coerce the participant to participate.
- In case of unconscious patients, the LAR can provide consent provided the investigator/LAR feels that the new intervention may yield good results.
- In case of a minor or unconscious person (noncompetent), the LAR can provide consent for such participants.
- The LAR should sign and date the informed consent document on behalf of the subject.
- The LAR should obtain a signature and date from the subject, and if the subject is illiterate, then a thumb impression can be obtained from them.[2]

What if the Legally Acceptable Representative is Illiterate?

Sometimes, even an LAR can be illiterate. In such a situation when the participant and his/her LAR are illiterate, an impartial witness may be chosen who is independent of the trial. This person should not be influenced by anyone (investigators/team members) involved with the trial and must accurately explain the details of the informed consent document. However, he/she can never consent on behalf of the subject as an LAR does. The impartial witness should sign and date the informed consent to document that he/she has fully explained the consent document to the participant. Audiovisual tools could be used in the consent process.[32]

OBJECTIVE 24.5. EXAMPLES OF SOME UNETHICAL PRACTICES IN CURRENT CLINICAL RESEARCH

Research involving humans is highly challenging as it involves the discovery of new interventions. Although ethical aspects are governed and established, there may be some unintentional concerns arising, which may lead to questionable results.

1. ***Disclosure of study medications to the patients***: If the trial is being carried out in a randomized, blinded crossover fashion, the disclosure of study medications to the researchers as well as to the patients may lead to the following consequences.

   Situation 1: The investigator/researcher who is interested in a positive outcome may be more likely to "see" these positive outcomes in the patients receiving the new intervention than in those patients receiving the placebo. It may, especially, be seen when the desired outcome of the study is "subjective" (e.g., when the primary objective of the clinical trial is the improvement in pain through changes in the visual analog scale).

   Situation 2: In the case of patients knowing that their study medication is a placebo, the intention to comply with the treatment procedure may diminish and may even force patients to withdraw their participation from the remaining periods. This may result in an unbalanced statistical approach.

   Situation 3: Patients who know that they are being treated with new interventional drugs may be psychologically influenced to feel an improvement in their health conditions, which sometimes may lead to falsification of results.

   To avoid these situations, it is important for the ethics committee to require that, whenever possible, a study is conducted only if the patient and the investigator are both "blinded" to the study intervention that a subject receives. Randomization of the treatment assignments also helps to reduce investigator bias, as does "blinding" the person analyzing as to who received the intervention versus who received the placebo if they are different from the investigator.
2. ***Underreporting of adverse events/serious adverse events***: All adverse events and serious adverse events occurring during a trial shall be reported at regular intervals to the ethics committee and sponsor as specified in the regulatory guidelines. Considering a multicenter study, there may be the occurrence of a serious adverse event such as anaphylaxis in only one center. The center may not report all the events as it is not happening in other centers, doubting the causal relationship. Underreporting of such occurrences of serious adverse events even though they occurred only in single study center to the ethics committee may lead to following consequences:
   - Imbalance in risk/benefit ratio.
   - Continuation of the study at the investigator's risk.
   - In either case, the ethics in safeguarding the well-being and health of patients are questioned.

   To avoid such scenarios, the ethics committee should conduct frequent monitoring visits and ensure that the adverse events are reported at regular frequencies.
3. ***Misrepresentation of data***: Scientific misconduct may take the form of fabrication or falsification of data or plagiarism. Generally, these are obvious attempts to misrepresent results. Occasionally, there can be an unintentional misrepresentation of results due to poor scientific design of an experiment.[16]

   Situation 1: The laboratory parameters will be assessed as described in the protocol at each visit after administration of the investigational product. The ethics committee should look at study design and analysis to ensure that changes in laboratory parameters are carefully evaluated to ensure accurate reporting of results and any changes in the laboratory parameters in each follow-up visit from the baseline screening values.

   For example, Patient X who is already hyperlipidemic receives an existing diabetic treatment in his first and third period and receives the new interventional diabetic drug treatment in his second and fourth period. Total cholesterol levels of the patient are shown in Table 24.1.

   There is an increase in the total cholesterol levels in Period 2 and Period 4. Is the new drug elevating this? An investigation that looks at cholesterol levels across the entire group would come to a different conclusion compared to examining the cholesterol levels based on treatment groups. Thus, it is important that the ethics committee carefully evaluates the scientific design to ensure appropriate analysis. Failing to do so at this stage may lead to

TABLE 24.1 Changes in Laboratory Parameters

Patient X	Period 1	Period 2	Period 3	Period 4
Total cholesterol Level	275 mg/dL	385 mg/dL	286 mg/dL	370 mg/dL

nonreporting of adverse events/reactions to the ethics committee, raising a concern. It also emphasizes that patients should be followed up at regular intervals factorizing their safety and health concern.

Situation 2: Prescribing information/investigator brochure/summary of product characteristics is a leaflet providing the details of the drug such as its indication, usage, dosage and administration, dosage forms and strengths, contraindications, warnings and precautions, adverse reactions, drug interactions, use in specific populations, overdosage, description, clinical pharmacology, nonclinical toxicology, clinical studies, references, storage, and patient counseling information.

The adverse reaction experienced with drug "X" is reported in such prescribing information/investigator brochure/summary of product characteristics. Such adverse events may be captured in the protocol and informed consent to make the participants aware of the occurrence of events during the conduct of the trial.

An adverse drug reaction mentioned in the product insert or summary of product characteristics could be missed in the informed consent/study protocol. The occurrence of such reaction in the trial is claimed to happen unexpectedly as it is missed out in the protocol/informed consent document. The reporting of the event by the investigator would be "unexpected" but actually, as per its label claim, the event is defined and occurrence is "expected." In such a case, the investigator may provide a wrong causality assessment to the ethics committee and the regulatory agencies.

Situation 3: Consider the same adverse reaction "upper respiratory tract infection" has been added to the protocol and informed consent in its amended version. The amended versions of informed consent should reach the site(s) and should be notified to the patients participating. The amended consent should be signed by all the participants. The amended informed consent document, which has "upper respiratory tract infection," should be appropriately informed to the study participants and be signed by them. If the amended consent document is not provided to the study participants, it may lead to a compromise in safety if the patient is prone to bronchitis or asthma attack.

4. ***Scientific misconduct***: Falsification of results or fabrication of data is the most common example of scientific misconduct. Data generated during the trial are essential and should be reported to the regulatory agencies as such. Any fabrication made to such data is highly unethical.[4]

Cases where fabrication may occur include the following:

Situation 1: Considering a hypertensive drug intervention study, the primary objective of the study would be to measure the decrease in the systolic and diastolic blood pressure in such patients.

The blood pressure values (source data) are subjective and can be easily fabricated. Monitoring of studies is essential for auditing the activities of the trial and for ensuring that the data generated are exactly recorded and reported at the end of the trial. The intention to falsify the data values is to provide positive results to the society, which is unacceptable.

Situation 2: Odd values should not be adjusted or ignored. It should be reported as such. Table 24.2 shows the hemoglobin levels of three subjects before and after the study intervention.

Subject B has their hemoglobin value reduced significantly at the end of the trial. The values obtained should be reported as such. The reason for a decrease in the value should be investigated. Appropriate treatment should be given, and the subject should be followed up until a significant increase in hemoglobin level is reached.

The values depicted in the table should be provided in the final report submitted to the regulatory agency. Any alteration of the originally generated out-of-range value to normal value is an offense, and moreover, such laboratory data should have their source data generated (which reflect the out-of-range value) from the hematological analyzer.

TABLE 24.2 Changes in Hemoglobin Levels Before and After the Study Period

Subjects	Hemoglobin Value Before Initiating the Study	Hemoglobin Value After Completion of the Study
A	11	9.5
B	14	7
C	10	8

Cross-verification of the data generated from the machine against the one submitted to the regulatory can be differentiated and questioned for any differences in future regulatory audits. This can help to identify potential misconduct and serves as a deterrent to temptations to either fabricate or falsify data.

5. ***Ethical assessment of industry-sponsored clinical trials***: Approximately 70% of the funding of clinical trials now come from pharmaceutical or biotechnology firms. It is not necessarily a bad thing, but given that there are commercial motivations driving such clinical trials, there are important considerations that must be carefully addressed to ensure that this source of funding does not bias the results that are being generated and published. Miller and Shorr report a case analysis where they identified important principles for evaluating industry-sponsored studies. The following seven areas need to be carefully evaluated by the ethics committee: (1) scientific value, (2) scientific validity, (3) fair subject selection, (4) favorable risk/benefit ratio, (5) independent review, (6) informed consent, and (7) respect for enrolled subjects.[33]

   Situation 1: Seeding trials are conducted by pharmaceutical companies in the name of research but are intended only for marketing purposes without any strong scientific objectives.[34] Various pharmaceutical companies have been warned by regulatory agencies for providing misleading information arising from this kind of seeding trials. Krumholz and colleagues provide a detailed dissection of the STEPS trial that was used to promote gabapentin and increase prescribing among investigators. The trial was planned and implemented by the marketing arm of the company. It was a phase IV uncontrolled, unblinded trial sponsored by the manufacturer. The stated objective was to study the efficacy, safety, tolerability, and quality of life among gabapentin users when titrating the drug to reduce the frequency of seizures. The authors of the original study did not mention their marketing objectives. However, the Krumholz's article identified the unstated marketing objective of the study to demonstrate efficacy, but an uncontrolled and unblinded study is a very poor design to demonstrate efficacy. They also used very complicated inclusion and exclusion criteria in the design, which would greatly decrease generalizability. The original IRB submission was denied twice because of these design deficiencies.[35]

   In an analysis of ADVANTAGE (Assessment of Differences between Vioxx And Naproxen To Ascertain Gastrointestinal Tolerability and Effectiveness), the trial identified the following characteristics of seeding trials:

   - there is marketing involved in study conception and design and in data collection and analysis and
   - there is nondisclosure of the study's true purpose from IRBs, patients, and investigators.[36]

   Situation 2: Sometimes pharmaceutical companies promote the marketing of drugs by providing information to the practicing physicians that reveal only the information that is important for the sale or use of a product, without providing full prescribing information. This "detailing" may come in the form of slick brochures, special "lunch and learn sessions," and continuing education presentations at exotic places. The influence of the pharmaceutical industry in altering prescribing patterns is studied.[37]

CONCLUSION

Ethics is an essential component of clinical research. Unethical studies have been conducted in the past and continue even today in different forms. Governments and other health agencies regulate research with relevant, updated laws and guidelines. However, conflicts of interest involving commercial products or services and the desire for research glory among researchers often get in the way of good science posing a challenge to good ethical practices. Therefore, ethics committees and their recommendations are vital to protect the well-being of research study participants. It also ensures that research involving human participants attains the highest quality standards.